

EXPERIMENT 6

DATA VISUALIZATION USING POWER BI

Aim

To import, transform, analyze, and visualize data using Power BI Desktop and create interactive dashboards and reports for decision-making.

Algorithm 1

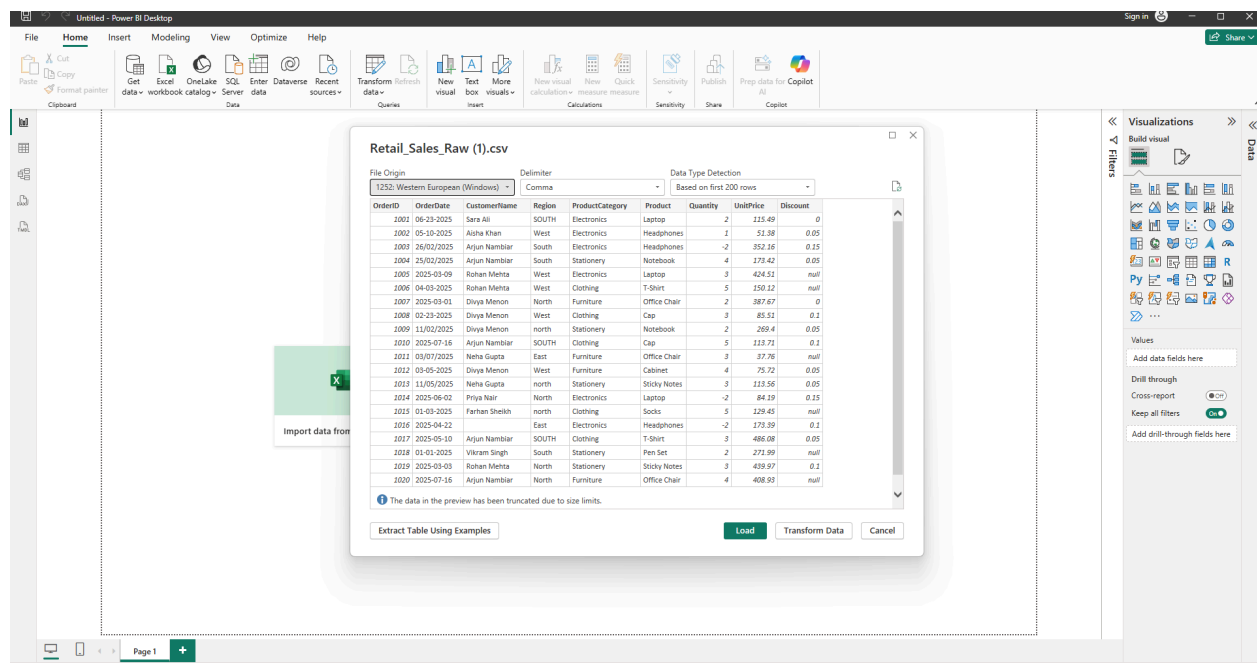
Importing Data into Power BI

Step 1 Open Power BI Desktop

Step 2 Select: Home → Get Data

Step 3 Choose CSV

Step 4 Browse dataset and select it



Algorithm 2

Data Transformation using Power Query

Step 1 Select Transform Data

Step 2 Open Power Query Editor

Step 3 Remove duplicates

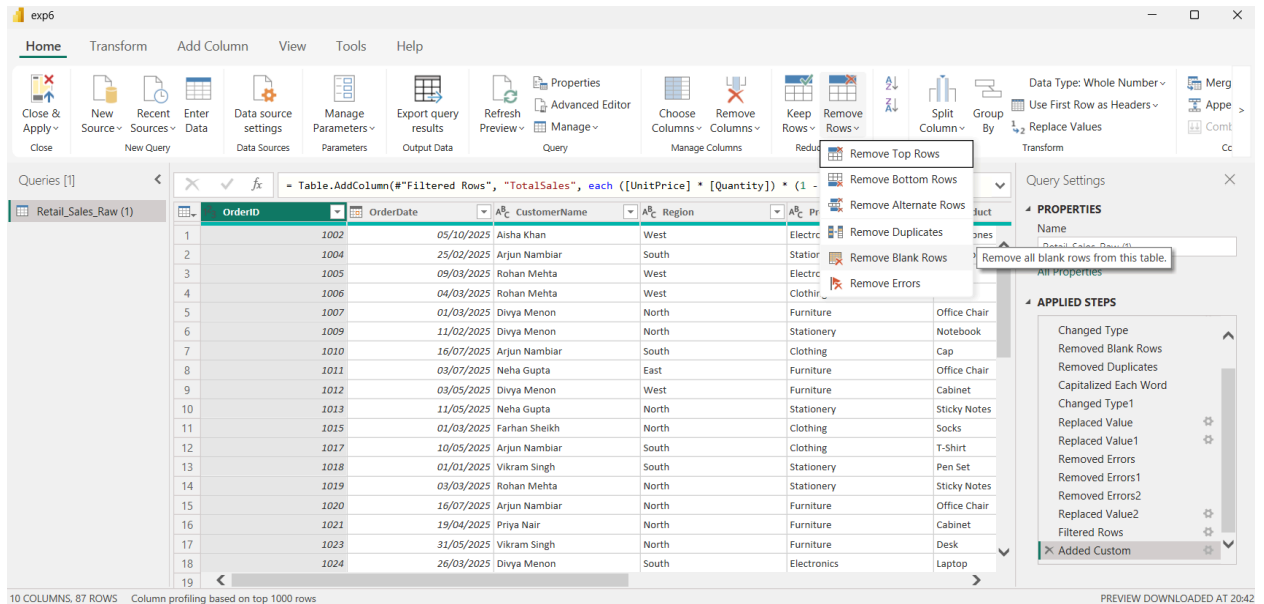
Step 4 Handle null values

Step 5 Rename columns

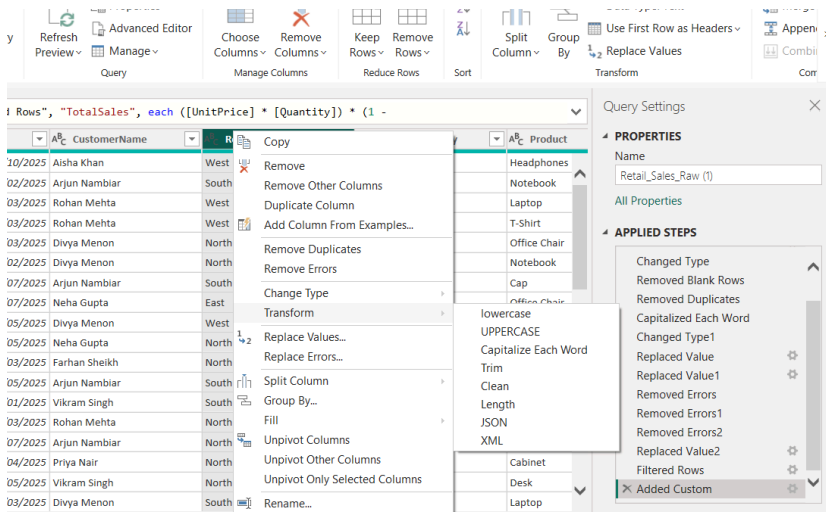
Step 6 Apply changes

Procedure

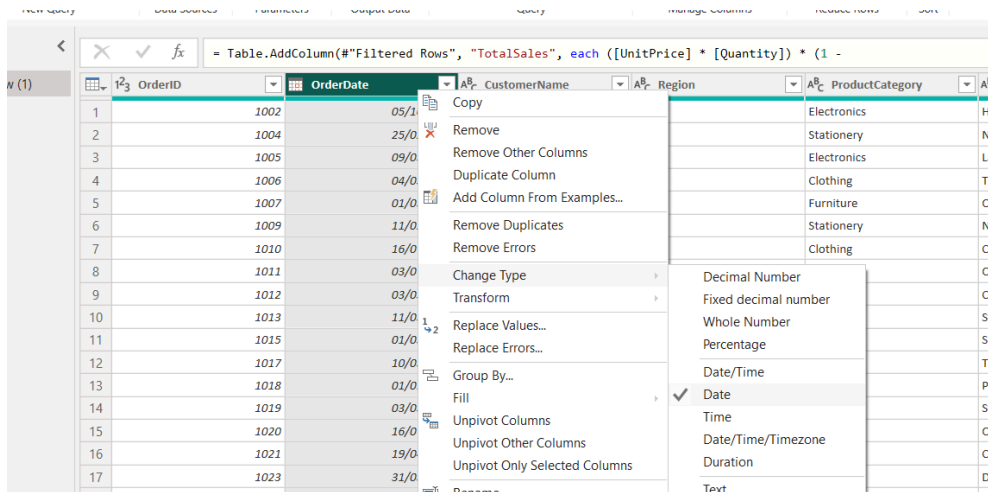
1. The data preview window will appear. Select Transform Data instead of Load to open the Power Query Editor.
2. In the Power Query Editor, examine the imported table and verify that all required columns are present, such as OrderID, OrderDate, CustomerName, Region, Quantity, UnitPrice, and Discount.
3. Remove blank rows from the dataset by selecting Home → Remove Rows → Remove Blank Rows. This removes completely empty records that could interfere with analysis and calculations.



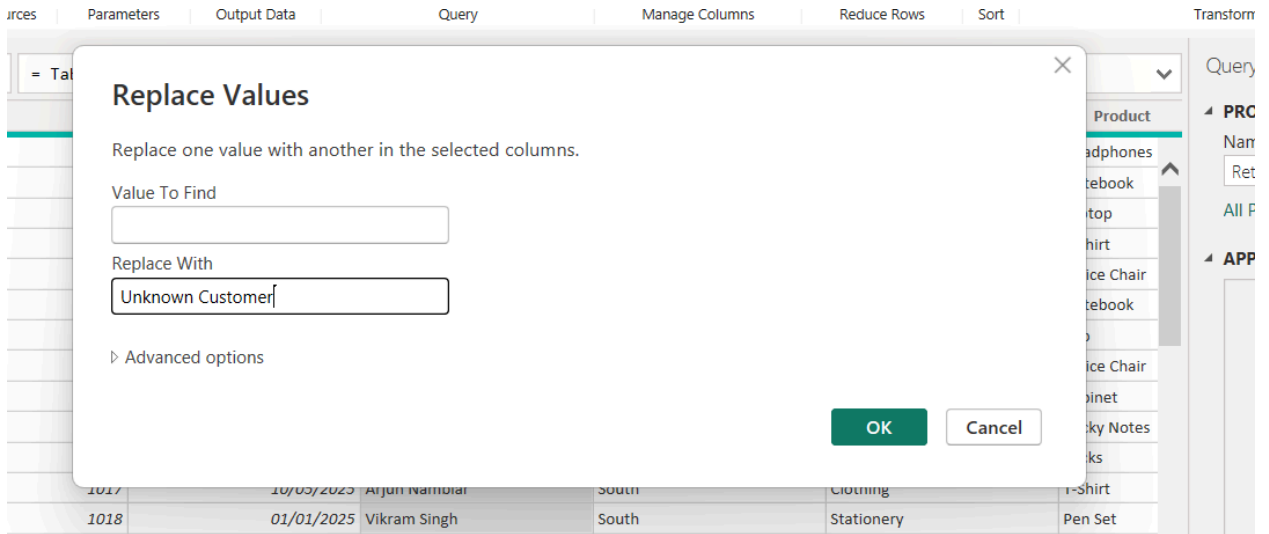
4. Remove duplicate records by selecting all columns in the table using Ctrl+A on the column headers and choosing Home → Remove Rows → Remove Duplicates. Power Query will remove records that contain identical values across all selected columns.
5. Standardize the values in the Region column. Right-click the Region column, select Transform, and choose Capitalize Each Word. This ensures that values such as “north”, “North”, and “NORTH” are represented consistently as “North”.



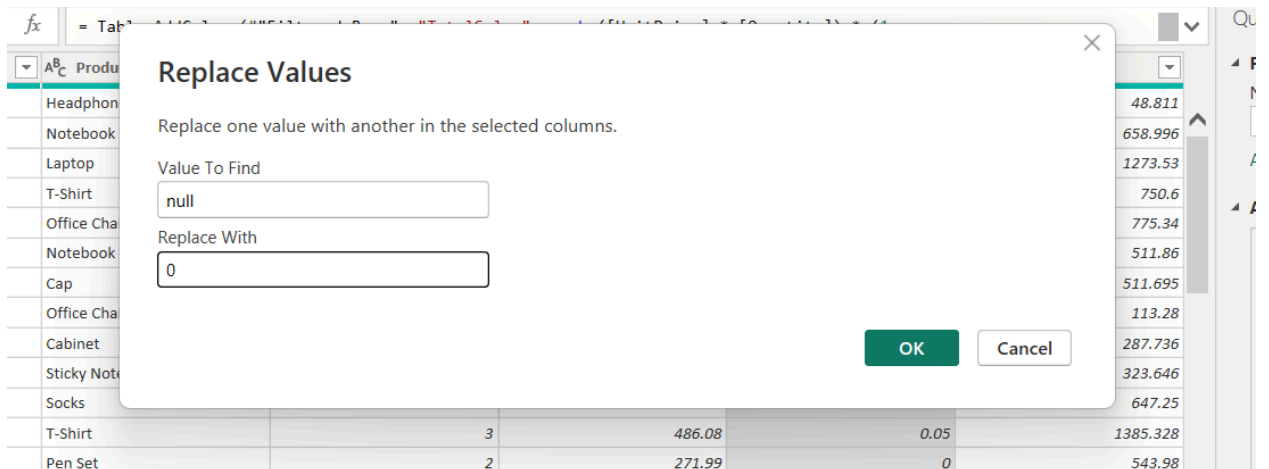
- Standardize the OrderDate column by selecting it and choosing Transform → Data Type → Date. If Power Query displays errors because the dates use an inconsistent format, right-click the OrderDate column, select Change Type → Using Locale, and choose the appropriate date format and locale.



- Handle missing values in the CustomerName column. Right-click CustomerName, select Replace Values, and replace blank or null values with “Unknown Customer”.



8. Handle missing values in the Discount column. Replace null values with 0 using the Replace Values option. A value of 0 represents that no discount was applied when the discount information is missing.



9. Identify and remove invalid Quantity values. Select the Quantity column and apply a filter by pressing the arrow next to the column name to exclude records where Quantity is less than 0. Negative quantities are treated as invalid business data because a normal sales transaction cannot contain a negative quantity.

Product	Quantity	UnitPrice	Discount
		51.38	0.05
		173.42	0.05
		424.51	0
		150.12	0
		387.67	0
		269.4	0.05
			0.1
			0
			0.05
			0.05
			0
			0.05
			0
			0.1

Filter Rows

Apply one or more filter conditions to the rows in this table.

Basic Advanced

Keep rows where 'Quantity'

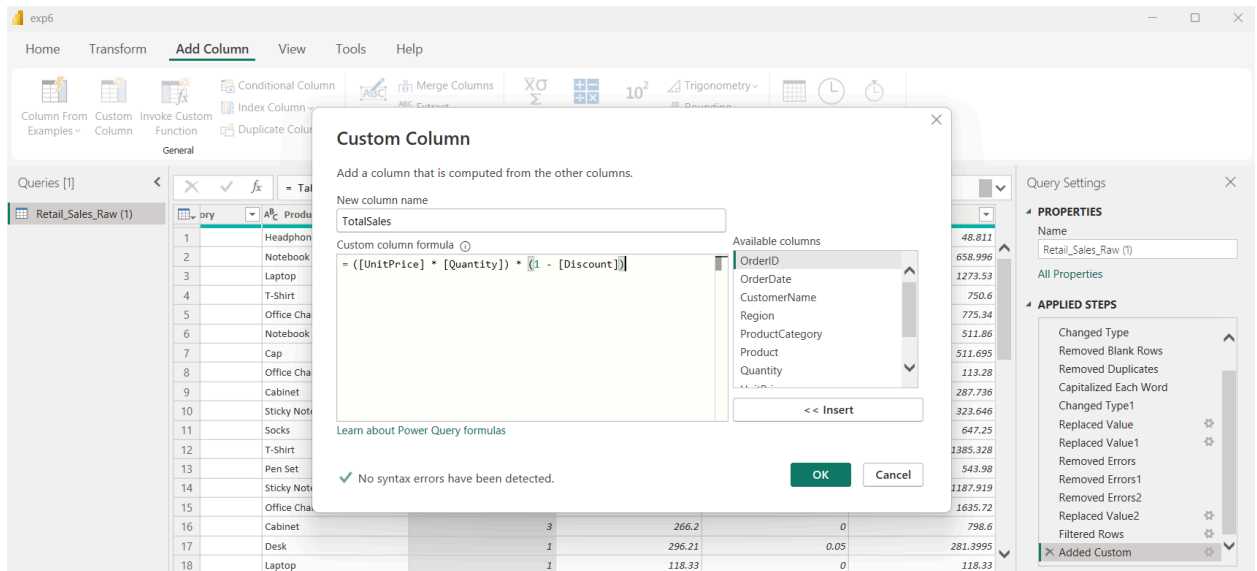
is greater than or equal to 0

And Or

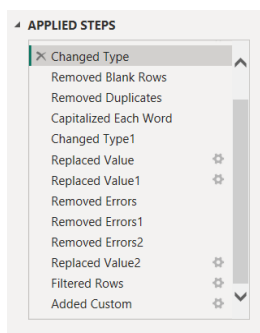
Enter or select a valu...

OK Cancel

10. Create a derived column for total sales. Select Add Column → Custom Column and name the new column TotalSales. Enter the formula ([UnitPrice] * [Quantity]) * (1 - [Discount]). This calculates the actual sales value after applying the discount.



11. Verify and correct the data types of all columns. Set OrderID to Whole Number, OrderDate to Date, Quantity to Whole Number, UnitPrice to Decimal Number, Discount to Decimal Number, TotalSales to Decimal Number, and the remaining descriptive columns to Text.
12. Review the Applied Steps pane on the right side of the Power Query Editor. Confirm that all transformation steps have been recorded in the correct order. If an error is present, select the corresponding step and correct the transformation.



13. After completing all cleaning and transformation operations, select Home → Close & Apply. Power BI will apply the transformations and load the cleaned data into the data model.
14. Switch to the Data/Table view and inspect the transformed table. Verify that duplicate rows, blank rows, inconsistent Region values, invalid dates, missing customer names, missing discounts, and negative quantities have been handled correctly.

exp6 • Last saved: Today at 11:02

File Home Help **Table tools** Share

Name: Retail_Sales_Raw (1)

Structure Relationships Calculations Mark as date table

OrderID	OrderDate	CustomerName	Region	ProductCategory	Product	Quantity	UnitPrice	Discount	TotalSales
1002	05 October 2025	Aisha Khan	West	Electronics	Headphones	1	51.38	0.05	49
1004	25 February 2025	Arjun Nambiar	South	Stationery	Notebook	4	173.42	0.05	659
1005	09 March 2025	Rohan Mehta	West	Electronics	Laptop	3	424.51	0	1274
1006	04 March 2025	Rohan Mehta	West	Clothing	T-Shirt	5	150.12	0	751
1007	01 March 2025	Divya Menon	North	Furniture	Office Chair	2	387.67	0	775
1009	11 February 2025	Divya Menon	North	Stationery	Notebook	2	269.4	0.05	512
1010	16 July 2025	Arjun Nambiar	South	Clothing	Cap	5	113.71	0.1	512
1011	03 July 2025	Neha Gupta	East	Furniture	Office Chair	3	37.76	0	113
1012	03 May 2025	Divya Menon	West	Furniture	Cabinet	4	75.72	0.05	288
1013	11 May 2025	Neha Gupta	North	Stationery	Sticky Notes	3	113.56	0.05	324
1015	01 March 2025	Farhan Sheikh	North	Clothing	Socks	5	129.45	0	647
1017	10 May 2025	Arjun Nambiar	South	Clothing	T-Shirt	3	486.08	0.05	1385
1018	01 January 2025	Vikram Singh	South	Stationery	Pen Set	2	271.99	0	544
1019	03 March 2025	Rohan Mehta	North	Stationery	Sticky Notes	3	439.97	0.1	1188
1020	16 July 2025	Arjun Nambiar	North	Furniture	Office Chair	4	408.93	0	1636
1021	19 April 2025	Priya Nair	North	Furniture	Cabinet	3	266.2	0	799
1023	31 May 2025	Vikram Singh	North	Furniture	Desk	1	296.21	0.05	281
1024	26 March 2025	Divya Menon	South	Electronics	Laptop	1	118.33	0	118
1025	28 May 2025	Farhan Sheikh	South	Furniture	Bookshelf	2	271.92	0	544
1026	01 April 2025	Karan Shah	West	Furniture	Cabinet	1	52.98	0.15	45
1028	19 April 2025	Karan Shah	South	Stationery	Notebook	2	270.46	0.05	514
1029	26 January 2025	Rohan Mehta	South	Stationery	Sticky Notes	4	404.97	0	1620
1030	13 April 2025	Priya Nair	South	Electronics	Laptop	4	245.39	0.05	932
1031	15 March 2025	Neha Gupta	North	Furniture	Cabinet	3	463.63	0.15	1182
1032	07 August 2025	Sara Ali	South	Clothing	Jacket	2	484.51	0	969
1033	10 February 2025	Aisha Khan	North	Electronics	Keyboard	5	241.01	0	1205
1034	14 April 2025	Imran Qureshi	North	Electronics	Mouse	1	339.23	0.05	322

Data: Search Retail_Sales_Raw (1)

15. To verify the ETL process visually, switch to Report view and create a Clustered Column Chart. Drag Region to the X- Axis field and TotalSales to the Y-axis field.

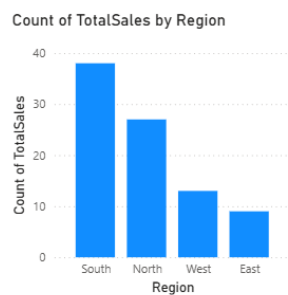
X-axis

Region

Y-axis

Count of TotalSales

16. Verify that the chart displays four standardized regions: North, South, East, and West. The resulting visual confirms that the data has been successfully extracted, transformed, and loaded into Power BI for further analysis.



Interpretation:

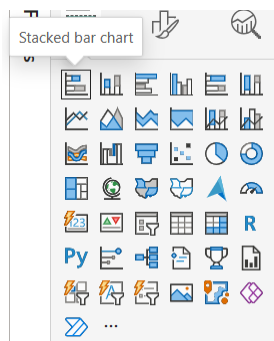
South region generated highest sales

East region generated lowest sales

Algorithm 3

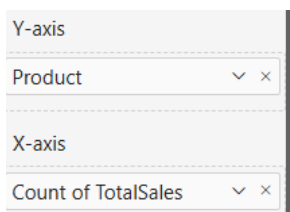
Creating a Bar Chart

Step 1 Choose Bar Chart Visualization



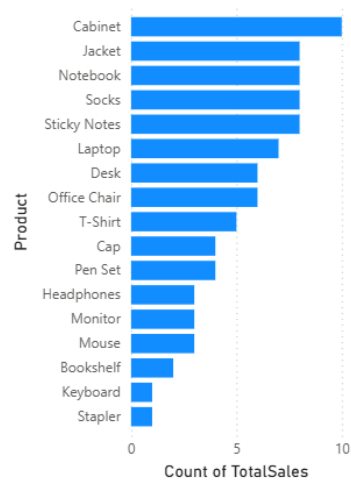
Step 2 Drag Product → Y-Axis

Step 3 Drag TotalSales → X axis



Step 4 Format chart

Count of TotalSales by Product



Interpretation:

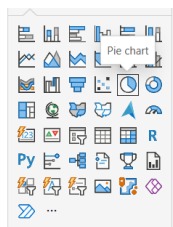
Cabinet generated highest sales

Stapler generated lowest sales

Algorithm 4

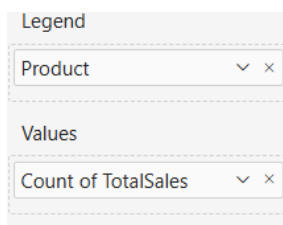
Creating a Pie Chart

Step 1 Select Pie Chart



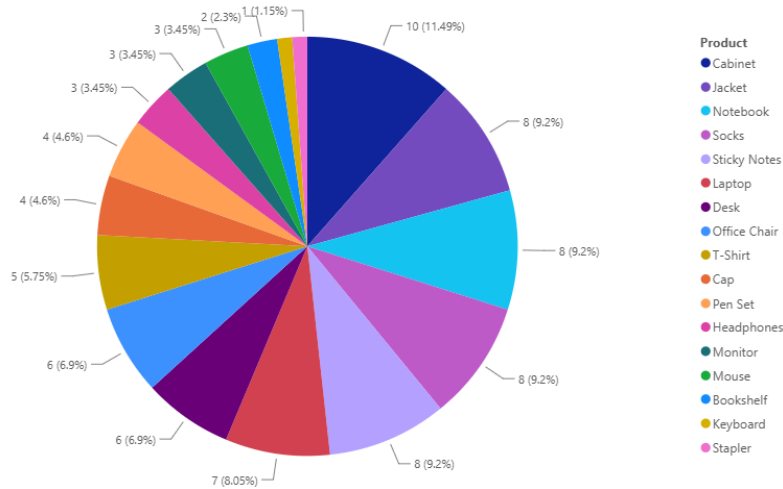
Step 2 Drag Product → Legend

Step 3 Drag TotalSales → Values



Output:

Count of TotalSales by Product



Interpretation:

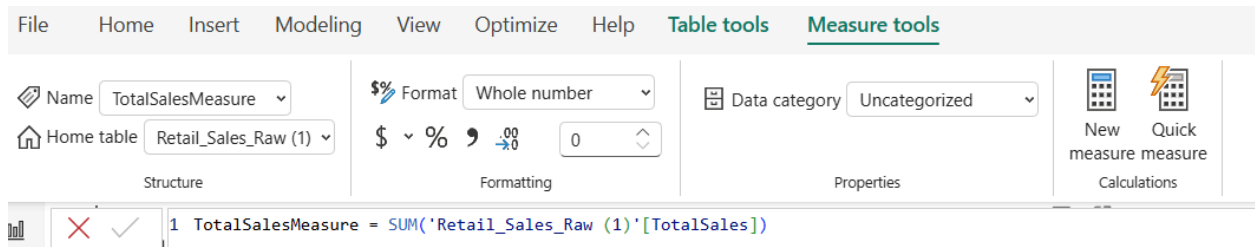
Cabinet generated highest Sales with 11.49%

Algorithm 5

Creating a KPI Card

Step 1 Create 4 measures by clicking on the New Measure from Measure tools tab, type the formula inside the dax field:

a. TotalSalesMeasure



b. Minimum

Name: Minimum Sales Home table: Retail_Sales_Raw (1)	Format: Whole number \$ % 0	Data category: Uncategorized	New measure Quick measure
Structure	Formatting	Properties	Calculations

1 Minimum Sales = MIN('Retail_Sales_Raw (1)')[TotalSales]

c. Maximum

Name: Maximum Sales Home table: Retail_Sales_Raw (1)	Format: Whole number \$ % 0	Data category: Uncategorized	New measure Quick measure
Structure	Formatting	Properties	Calculations

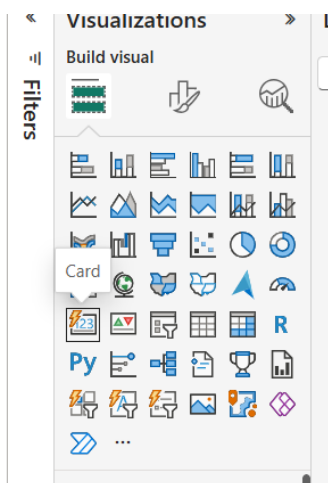
1 Maximum Sales = MAX('Retail_Sales_Raw (1)')[TotalSales]

d. Average

Name: AverageSalesMeas... Home table: Retail_Sales_Raw (1)	Format: General \$ % Auto	Data category: Uncategorized	New measure Quick measure
Structure	Formatting	Properties	Calculations

1 AverageSalesMeasure = AVERAGE('Retail_Sales_Raw (1)')[TotalSales]

Step 2 Create card visual

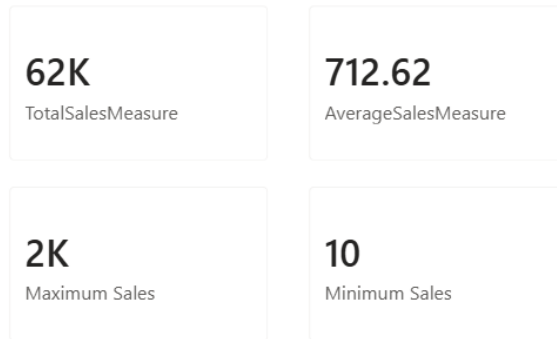


Step 3 Drag that measure into the field

Value

TotalSalesMeasure ▼ ×

Output:

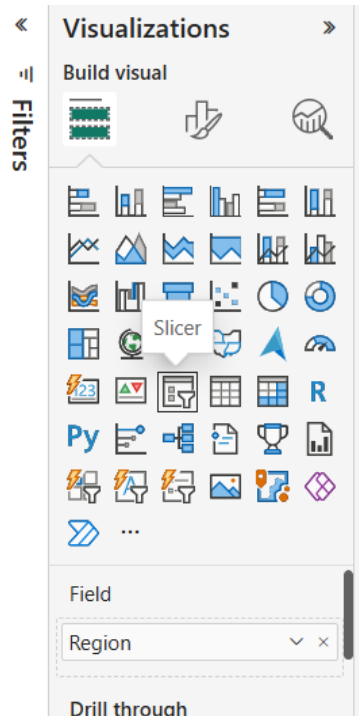


Algorithm 6

Creating Slicers

Step 1 Select Slicer Visualization

Step 2 Drag Region Field



Step 3 Enable Single Selection

Step 4 Apply Filters

Output:

Without any filter:

Region
☐ East
☐ North
☐ South
☐ West

62K

TotalSalesMeasure

712.62

AverageSalesMeasure

2K

Maximum Sales

10

Minimum Sales

With north filter:

Region
☐ East
☒ North
☐ South
☐ West

19K

TotalSalesMeasure

712.89

AverageSalesMeasure

2K

Maximum Sales

29

Minimum Sales

Algorithm 8

Creating Dashboard

Step 1 Add KPI Card

Step 2 Add Bar Chart

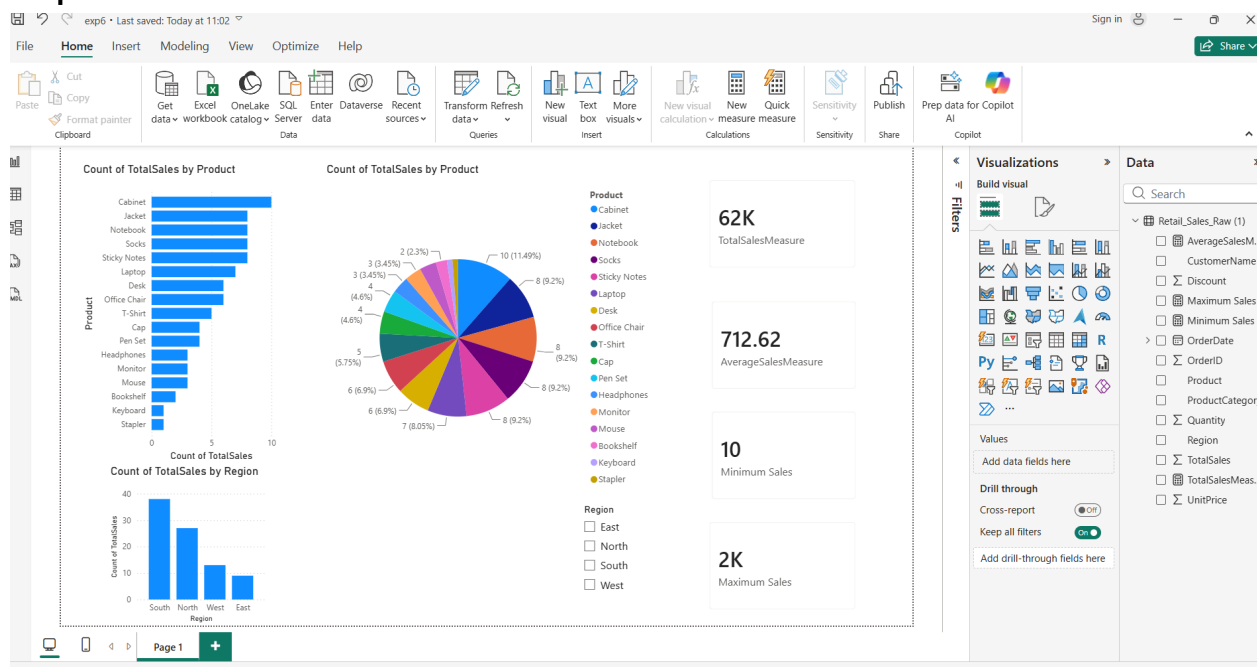
Step 3 Add Pie Chart

Step 4 Add Slicer

Step 5 Arrange Visuals

Step 6 Save Dashboard

Output:



Result:

The data was successfully imported, transformed, analyzed, and visualized in Power BI Desktop, and an interactive dashboard was created for effective decision-making.